

EEG-Driven Image Reconstruction with Saliency-Guided Diffusion Models

Igor Abramov

ig.abramov@innopolis.university

Ivannikov Institute for System Programming of the Russian Academy of Sciences

Research Center for Trusted Artificial Intelligence

Moscow, Russia

ITMO University

AI Talent Hub

Saint Petersburg, Russia

Abstract

Existing EEG-driven image reconstruction methods often overlook spatial attention mechanisms, limiting fidelity and semantic coherence. To address this, we propose a dual-conditioning framework that combines EEG embeddings with spatial saliency maps to enhance image generation. Our approach leverages the Adaptive Thinking Mapper (ATM) for EEG feature extraction and fine-tunes Stable Diffusion 2.1 via Low-Rank Adaptation (LoRA) to align neural signals with visual semantics, while a ControlNet branch conditions generation on saliency maps for spatial control. Evaluated on THINGS-EEG, our method achieves a significant improvement in the quality of low- and high-level image features over existing approaches. Simultaneously, strongly aligning with human visual attention. The results demonstrate that attentional priors resolve EEG ambiguities, enabling high-fidelity reconstructions with applications in medical diagnostics and neuroadaptive interfaces, advancing neural decoding through efficient adaptation of pre-trained diffusion models.

CCS Concepts

• Computing methodologies → Reconstruction.

Keywords

Multimodal Interaction, EEG, Diffusion Models, ControlNet, LoRA, Image Reconstruction, BCI

ACM Reference Format:

Igor Abramov. 2025. EEG-Driven Image Reconstruction with Saliency-Guided Diffusion Models. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3754476>

1 Introduction

The integration of neural data with generative models has emerged as a promising direction for brain-computer interfaces and cognitive computing. Within this domain, decoding visual experiences

from brain activity remains a fundamental challenge [9, 19]. While fMRI-based stimulus reconstruction shows promise [6, 26], EEG offers greater practicality through portability and temporal resolution [27]. However, EEG's low signal-to-noise ratio has historically limited most work to classification rather than pixel-level synthesis.

Recent advances provide new pathways: The Adaptive Thinking Mapper (ATM) generates EEG embeddings aligned with visual semantics [16], while diffusion models [23] and ControlNet [28] enable high-fidelity conditional image generation. Large-scale neuroimaging datasets (THINGS-EEG [4], EEG-ImageNet [30]) have further accelerated progress. Crucially, existing approaches overlook visual attention patterns despite their perceptual importance [2, 5, 8] and potential for resolving EEG ambiguities [29].

Saliency maps predict human visual attention patterns, offering interpretability for computer vision and conditioning signals for generative systems [24]. Early approaches used bottom-up statistical features [2, 5, 8], while modern methods leverage deep learning and eye-tracking datasets (CAT2000 [1], MIT1003 [12], SALICON [11]) to predict low-level and semantic attention [10, 14, 21]. GazeFusion [29] demonstrated saliency-guided diffusion models, reflecting how attention combines low-level features and high-level semantics [15].

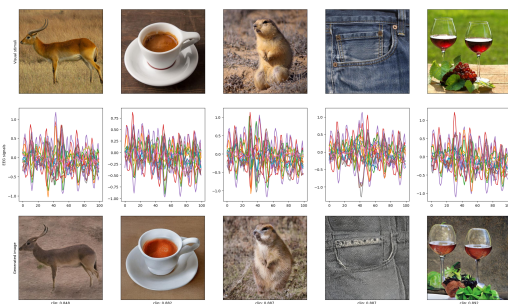


Figure 1: Example stimuli and reconstructions showing original images (top), corresponding EEG signals (middle), and our model's outputs (bottom) conditioned on both EEG and saliency patterns.

We bridge this gap with a novel dual-conditioning approach combining: 1) Semantic alignment of EEG embeddings via LoRA [7], and 2) Spatial guidance through ControlNet using predicted saliency maps. This integration leverages both semantic content

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '25, Dublin, Ireland

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3754476>

and human-like attention patterns, significantly improving reconstruction quality (Fig. 1) across pixel-level, structural, and semantic metrics versus EEG-only baselines. Implementation details are available at GitHub¹.

2 Proposed Framework

Our framework integrates EEG-conditioned image generation with spatial attention control through a novel dual-conditioning approach, as illustrated in Figure 2. The system combines EEG feature extraction, latent diffusion modeling, and saliency-guided control in a multi-stage pipeline.

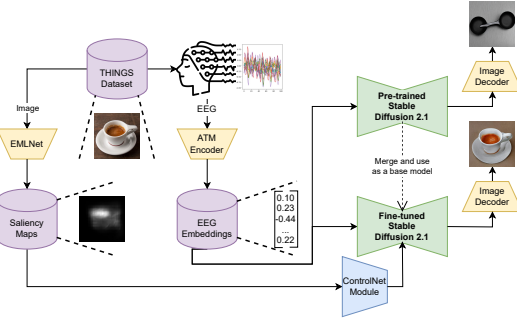


Figure 2: Workflow of our EEG and saliency-conditioned image generation framework. Stage 1: LoRA fine-tuning of Stable Diffusion with EEG embeddings. Stage 2: ControlNet training with saliency maps while keeping the EEG-conditioned model frozen.

We employ the Adaptive Thinking Mapper (ATM) encoder [16] to process raw EEG signals from the THINGS-EEG dataset [4]. The architecture preserves EEG’s spatial and temporal features using channel-wise attention and specialized convolutions, producing meaningful embeddings that match visual concepts.

Building on Stable Diffusion 2.1 [23], we adapt the image generation process to accept EEG embeddings as semantic conditioning. Using Low-Rank Adaptation (LoRA) [7], we efficiently fine-tune the cross-attention layers of the UNet to respond to neural patterns while preserving the model’s original generative capabilities. We trained this setup on single RTX4080 in half precision. AdamW [18] with $\text{lr} = 2e^{-3}$ and default betas performed 212000 optimization steps with batch size 8. We also used Cosine Annealing Warm Restarts [17] lr scheduler with $\text{eta_min} = 1e^{-6}$, $T_0 = 5000$.

The final enhancement incorporates ControlNet [28] to condition generation on predicted saliency maps from EMLNet [10]. This spatial conditioning branch directs image composition according to human attention patterns while maintaining the semantic content derived from EEG. We trained this setup on single RTX4080 in full precision. AdamW [18] with $\text{lr} = 1e^{-4}$ and default betas performed 65000 optimization steps with batch size 2 and 4 gradient accumulation steps. We also used Cosine Annealing Warm Restarts [17] lr scheduler with $\text{eta_min} = 1e^{-5}$, $T_0 = 5000$.

As shown in Figure 1, the combined conditioning produces images that align with both the conceptual and attentional aspects of human vision.

¹<https://github.com/IGragon/EEG-Saliency-Image>

3 Experimental Results

We evaluate our dual-conditioning approach using reconstruction metrics (low-level and semantic) and saliency metrics. Quantitative comparisons use Subject 8 data from THINGS-EEG [4] against EEG-based image reconstruction framework proposed in [16] with different EEG encoders.

Tables 1 and 2 demonstrate significant improvements. Our saliency-guided approach achieves a significantly higher improvement in pixel correlation (PixCorr) and structure similarity index (SSIM) over the [16] framework. Semantic metrics (i.e. [3, 13, 22, 25] using [20]) show near-perfect scores, confirming superior preservation of both low-level features and high-level semantics.

Table 1: Low-level reconstruction metrics (Subject 8). Higher is better.

Approach	PixCorr $\uparrow$	SSIM $\uparrow$
VDaR, ATM [16]	0.160	0.345
Ours: EEG-only	0.080	0.271
Ours: Saliency-guided	0.473	0.369

Table 2: High-level reconstruction metrics (Subject 8).

Approach	AlexNet (2) $\uparrow$	AlexNet (5) $\uparrow$	Inception $\uparrow$	CLIP $\uparrow$	SwAV $\downarrow$
VDaR, ATM [16]	0.776	0.866	0.734	0.786	0.582
Ours: EEG-only	0.774	0.865	0.745	0.767	0.593
Ours: Saliency-guided	0.999	0.998	0.946	0.904	0.453

Table 3 confirms our spatial conditioning significantly improves attention alignment. Saliency control yields higher correlation coefficient (CC), lower KL divergence, and higher SIM compared to EEG-only conditioning.

Table 3: Saliency metrics (Subject 8).

Approach	CC $\uparrow$	KL $\downarrow$	SIM $\uparrow$
EEG-only	0.51	2.99	0.60
Saliency-guided	0.85	0.52	0.80

4 Conclusion

We presented a novel EEG-conditioned image generation framework enhanced with saliency guidance, demonstrating significant improvements in both reconstruction quality and attention alignment. Our approach achieves noticeable performance with higher pixel correlation and better saliency correlation compared to EEG-only baselines, validating that spatial attention cues resolve ambiguities in neural decoding. The work establishes that parameter-efficient adaptation of diffusion models can effectively incorporate both EEG embeddings and saliency maps, opening new possibilities for brain-computer interfaces. Future directions include EEG-predicted saliency estimation and cross-subject generalization to advance practical applications in cognitive neuroscience and assistive technologies.

Acknowledgments

The work was supported by a grant, provided by the Ministry of Economic Development of the Russian Federation in accordance with the subsidy agreement (agreement identifier 000000C313925P4G0002) and the agreement with the Ivannikov Institute for System Programming of the Russian Academy of Sciences dated June 20, 2025 No. 139-15-2025-011.

References

- [1] Ali Borji and Laurent Itti. 2015. CAT2000: A Large Scale Fixation Dataset for Boosting Saliency Research. doi:10.48550/ARXIV.1505.03581
- [2] Neil Bruce and John Tsotsos. 2005. Saliency Based on Information Maximization. In *Advances in Neural Information Processing Systems*, Y. Weiss, B. Schölkopf, and J. Platt (Eds.), Vol. 18. MIT Press. https://proceedings.neurips.cc/paper_files/paper/2005/file/0738069b244a1c43c83112b735140a16-Paper.pdf
- [3] Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. 2020. Unsupervised Learning of Visual Features by Contrasting Cluster Assignments. doi:10.48550/ARXIV.2006.09882
- [4] Alessandro T. Gifford, Kshitij Dwivedi, Gemma Roig, and Radosław M. Cichy. 2022. A large and rich EEG dataset for modeling human visual object recognition. *NeuroImage* 264 (Dec. 2022), 119754. doi:10.1016/j.neuroimage.2022.119754
- [5] Jonathan Harel, Christof Koch, and Pietro Perona. 2006. Graph-Based Visual Saliency. In *Advances in Neural Information Processing Systems*, B. Schölkopf, J. Platt, and T. Hoffman (Eds.), Vol. 19. MIT Press. https://proceedings.neurips.cc/paper_files/paper/2006/file/4db0f8b0fc895da263fd77fc8aeca4e4-Paper.pdf
- [6] Jun Kai Ho, Tomoyasu Horikawa, Kei Majima, Fan Cheng, and Yukiyasu Kamitani. 2023. Inter-individual deep image reconstruction via hierarchical neural code conversion. *NeuroImage* 271 (May 2023), 120007. doi:10.1016/j.neuroimage.2023.120007
- [7] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=nZeVKeeFYf9>
- [8] Laurent Itti and Christof Koch. 2001. Computational modelling of visual attention. *Nature Reviews Neuroscience* 2, 3 (March 2001), 194–203. doi:10.1038/35058500
- [9] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yunhsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision. doi:10.48550/ARXIV.2102.05918
- [10] Sen Jia and Neil D.B. Bruce. 2020. EML-NET: An Expandable Multi-Layer NET-work for saliency prediction. *Image and Vision Computing* 95 (2020), 103887. doi:10.1016/j.imavis.2020.103887
- [11] Ming Jiang, Shengsheng Huang, Juanyong Duan, and Qi Zhao. 2015. SALICON: Saliency in Context. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 1072–1080. doi:10.1109/CVPR.2015.7298710
- [12] Tilke Judd, Krista Ehinger, Frédo Durand, and Antonio Torralba. 2009. Learning to predict where humans look. In *2009 IEEE 12th International Conference on Computer Vision*. 2106–2113. doi:10.1109/ICCV.2009.5459462
- [13] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. 2012. ImageNet Classification with Deep Convolutional Neural Networks. In *Advances in Neural Information Processing Systems*, F. Pereira, C.J. Burges, L. Bottou, and K.Q. Weinberger (Eds.), Vol. 25. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf
- [14] Matthias Kümmerer, Matthias Bethge, and Thomas S. A. Wallis. 2022. DeepGaze III: Modeling free-viewing human scanpaths with deep learning. *Journal of Vision* 22, 5 (April 2022), 7. doi:10.1167/jov.22.5.7
- [15] Matthias Kümmerer, Thomas S. A. Wallis, and Matthias Bethge. 2015. Information-theoretic model comparison unifies saliency metrics. *Proceedings of the National Academy of Sciences* 112, 52 (Dec. 2015), 16054–16059. doi:10.1073/pnas.1510393112
- [16] Dongyang Li, Chen Wei, Shiyang Li, Jiachen Zou, Haoyang Qin, and Quanying Liu. 2024. Visual Decoding and Reconstruction via EEG Embeddings with Guided Diffusion. arXiv:2403.07721 [cs.HC] <https://arxiv.org/abs/2403.07721>
- [17] Ilya Loshchilov and Frank Hutter. 2016. SGDR: Stochastic Gradient Descent with Warm Restarts. doi:10.48550/ARXIV.1608.03983
- [18] Ilya Loshchilov and Frank Hutter. 2017. Decoupled Weight Decay Regularization. doi:10.48550/ARXIV.1711.05101
- [19] Yoichi Miyawaki, Hajime Uchida, Okito Yamashita, Masa-aki Sato, Yusuke Morito, Hiroki C. Tanabe, Norihiro Sadato, and Yukiyasu Kamitani. 2008. Visual Image Reconstruction from Human Brain Activity using a Combination of Multiscale Local Image Decoders. *Neuron* 60, 5 (Dec. 2008), 915–929. doi:10.1016/j.neuron.2008.11.004
- [20] Furkan Ozelcelik and Rufin VanRullen. 2023. Natural scene reconstruction from fMRI signals using generative latent diffusion. *Scientific Reports* 13, 1 (Sept. 2023). doi:10.1038/s41598-023-42891-8
- [21] Junting Pan, Cristian Canton Ferrer, Kevin McGuinness, Noel E. O'Connor, Jordi Torres, Elisa Sayrol, and Xavier Giro-i Nieto. 2017. SalGAN: Visual Saliency Prediction with Generative Adversarial Networks. doi:10.48550/ARXIV.1701.01081
- [22] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. doi:10.48550/ARXIV.2103.00020
- [23] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2021. High-Resolution Image Synthesis with Latent Diffusion Models. arXiv:2112.10752 [cs.CV]
- [24] Karolina Szczepankiewicz, Adam Popowicz, Kamil Charkiewicz, Katarzyna Nałęcz-Charkiewicz, Michał Szczepankiewicz, Sławomir Lasota, Paweł Zawistowski, and Krystian Radlak. 2023. Ground truth based comparison of saliency maps algorithms. *Scientific Reports* 13, 1 (Oct. 2023). doi:10.1038/s41598-023-42946-w
- [25] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. 2015. Rethinking the Inception Architecture for Computer Vision. doi:10.48550/ARXIV.1512.00567
- [26] Yu Takagi and Shinji Nishimoto. 2023. High-resolution image reconstruction with latent diffusion models from human brain activity. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 14453–14463. doi:10.1109/CVPR52729.2023.01389
- [27] Francis R. Willett, Donald T. Avansino, Leigh R. Hochberg, Jaimie M. Henderson, and Krishna V. Shenoy. 2021. High-performance brain-to-text communication via handwriting. *Nature* 593, 7858 (May 2021), 249–254. doi:10.1038/s41586-021-03506-2
- [28] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding Conditional Control to Text-to-Image Diffusion Models.
- [29] Yunxiang Zhang, Nan Wu, Connor Z Lin, Gordon Wetzstein, and Qi Sun. 2024. GazeFusion: Saliency-guided Image Generation. *ACM Transactions on Applied Perception* (2024).
- [30] Shuqi Zhu, Ziyi Ye, Qingyao Ai, and Yiqun Liu. 2024. EEG-ImageNet: An Electroencephalogram Dataset and Benchmarks with Image Visual Stimuli of Multi-Granularity Labels. doi:10.48550/ARXIV.2406.07151